

1. **Prompt:** I want to pick the most ambiguous and difficult problem out of the 4 mentioned in this assignment. I have 15 working hours to solve any one of these. Rate these 4 problems on ambiguity, technical difficulty and doability and give me a recommendation on which problem to pick.

Problem	Ambiguity	Technical difficulty	Doability (15h)	
A — Investment Committee Multi-agent debate + token budget	 6/10	 7/10	 7/10	solid
B — Deterministic Agent Path locking + coding agent	 9/10	 9/10	 4/10	risky
C — Contradicting Memory Belief system + conflict resolution	 8/10	 8/10	 6.5/10	recommended
D — Decision Graph Knowledge graph + NL extraction	 5/10	 6/10	 8/10	easiest

Picked problem 2 as it seemed to be the right mix of technical execution challenge and ambiguity.

2. **Prompt:** Let's design this system and then we will go down into details. Let's try to get a system design document in place. Put down all the functional as well as non-functional specifications for this system, followed by high level components, diagram, considerations and trade-offs etc. like a proper faang staff engineer level system design document.

System design document 1 generated

3. Prompt (Follow up to Prompt 2):

answer the following doubts:

1. does it make sense to cache results at every say checkpoint so that the tokens and compute used is conserved, which means there will have to be criterias to match against to decide if the cached results can be used or fresh rerun for all stages is needed.
2. Out of all the decision variations such as fast api vs cli, which ones are a sort of no-op, which we know won't make a difference to output or will make minimal output, quantify this across all the decision alternatives in the problem so that we will focus effort on entertaining more variance in some layers and keep things same in others such as choose just one out of numpy, pandas etc. if the variance will be low
3. introduce caches, redis as well as in-memory
4. introduce max cost per run defaulting to some value, if the cost goes beyond the run halts, introduce a cost estimator as well that does it upfront and asks user if they want to go though still in cli

CATEGORY	DECISION POINT	OUTPUT VARIANCE (0-10)	TIER
Algorithm	Anomaly detection method		9.5 High
Data	Ticker selection		9.0 High
Data	Timeframe		8.0 High
Algorithm	Anomaly parameters		7.5 High
Data	Data fields (OHLCV vs close)		7.0 Medium
Architecture	Report format		6.5 Medium
Recovery	Error recovery strategy		6.0 Medium
Library	Visualization library		5.0 Medium
Architecture	Project structure		3.5 Low
Library	Data library (pandas/polars)		2.0 Low
Architecture	API framework		1.5 No-op
Optimization	Caching strategy		1.0 No-op
Architecture	Error handling pattern		1.0 No-op

Variance = how much the final output (anomalies found, report content, API response shape) changes

System design document 2 generated

4. Follow up to prompt 3 above with Jnara guidelines and complete problem statement as context:

given the problem and guidelines in the attached document and above system design, give me a thorough build plan that achieves must do as well as stretch goals. Give me a detailed step-by step document with what we will build in each step, what boxes it will tick in terms of goals and guidelines in the document, along with prompt i should give claude to build that piece including description of tests and manual tests I should do to verify core functionalities anyways for each step. Assume I want to build a containerized service which will have redis, postgres, the python service altogether so it is easy to develop and test locally. Make sure to direct the ai ide to follow python coding best practices, interfaces and design patterns and keep env files for all sorts of config.

Build plan version 1 generated

5. Follow up to prompt 4 above:

give me suggestions for llm providers such that i can use models across the board with just one api key, then give me decision making matrix on how to identify which model to use for which task, focus on minimizing token usage without compromising work quality. don't have bias for claude models, give me reason why you are choosing that specific model

Build plan version 2 generated

6. Fresh prompt with latest version of system design and build plan attached for context:

implement phase 0 as mentioned in the build plan. Refer rest of the build plan or sdd for high level understanding if needed. both in the attached documents. Refer the main branch code for any deviations from the plan to fix issues missed in the plan.

The above prompt repeated for each phase followed by running tests and manual verification scenarios for each phase, merging into main and continuing the same for the next phase. For issues found in any phase feature branch, a corresponding patch/ branch created for the fix and merged post verification.